

Evaluating and Equivalent Expressions*Question Bank - Additional 3.06*

Name _____

Date _____

Period _____

Show your work. Circle your final answer.

For questions 1 through 5, apply the laws of exponents to evaluate. Show your steps.

1. $\frac{4^{-3}}{4^{-5}} \cdot 4^4$

2. $\frac{((-5)^3)^4}{(-5)^2 \cdot (-5)^{-3}}$

3. $\left(\left(\frac{2}{3}\right)^0 \cdot \left(\frac{3}{2}\right)^{-2}\right)^3$

4. $\frac{\left(\frac{1}{3}\right)^2 \cdot \left(\frac{1}{3}\right)^3}{\left(\frac{1}{3}\right)^{-4}}$

$$5. \quad \left(\frac{2^3 \cdot 2^4}{2^{-3}} \right)^{-2}$$

Evaluate each expression in questions 6 through 9. Show all the steps.

$$6. \quad \frac{\left(\frac{1}{3}\right)^{-5} \cdot (3^7)^0}{(3^2)^{-3}}$$

$$7. \quad \frac{2 \cdot 3^{-4} \cdot 2 \cdot 3^0}{(2 \cdot 3^2)^{-2}}$$

$$8. \quad \frac{\left[\left(\frac{1}{4}\right)^{-5} \cdot \left(\frac{1}{4}\right)^2\right]^2}{\left(\frac{1}{4}\right)^{-4}}$$

$$9. \quad \frac{(4^{-1})^3 \cdot \left(\frac{1}{4}\right)^{-4}}{4^3}$$

Common bases.

10. Write each expression as a single power with the base given.

a. 81^3 with base 3

b. 64^3 with base 2

c. $4^2 \cdot 64^3$ with base 4

d. $27 \cdot 3^{-5}$ with base 3

11. Paul says $\left(\frac{1}{8}\right)^2 \cdot 2^{-2}$ and $\left(\frac{1}{4}\right)^4 \cdot 4^{-1}$ are equivalent because both can be written as 2^{-8} .

a. Write the first expression as a single power of 2. Show your work.

b. Write the second expression as a single power of 2. Show your work.

c. Is Paul correct?

12. Which expression is equivalent to $\frac{3^5}{6^2}$?

- ☐ A. $\frac{3^3}{2^2}$
- ☐ B. $\left(\frac{3}{6}\right)^3$
- ☐ C. 3^3
- ☐ D. $\frac{3^7}{2^2}$

13. Find the value of n that makes each statement true.

a. $27^n = 3^{12}$

b. $\left(\frac{1}{4}\right)^n = 2^{10}$